Rates and Best Buys

Name _____ Date _____ Calculator allowed Version 1

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KEY WORDS rate *(pāpātanga)* · unit rate *(pāpātanga kotahi)* · best buy · speed *(tere)*

A Unit rates ●○○

Remember: A rate compares two different quantities. "Per" means "for each one".

1. Find each rate.

a) $12 for 4 kg: cost per kg = _____

b) 240 km in 3 hours: km per hour = _____

c) $45 for 5 hours of work: pay per hour = _____

d) 600 words in 8 minutes: words per minute = _____

e) 18 L in 4 minutes: litres per minute = _____

f) $7.20 for 6 bottles: cost per bottle = _____

g) 360 heartbeats in 5 minutes: beats per minute = _____

h) $2.80 for 400 g: cost per 100 g = _____

B Using a rate ●○○

2. Answer each question.

a) A tap runs at 12 litres per minute. How much water runs out in 7 minutes?

b) Mere earns $24.50 per hour. How much does she earn in 6 hours?

c) A car uses 8 litres of petrol for every 100 km. How much petrol does it use on a 350 km trip?

d) A printer prints 24 pages per minute. How long does it take to print 300 pages?

e) Mobile data costs $15 per GB. Find the cost of 2.5 GB.

f) A heart beats 72 times per minute. How many times does it beat in 1 hour?

C Best buys ●●○

Remember: Find the price of one unit for each deal (the unit price), then compare.

3. Which is the better buy? Show the unit price for each.

a) Rice: 2 kg for $5.60 or 5 kg for $13.50

b) Juice: 1.5 L for $3.90 or 2 L for $5

c) Pens: 6 pens for $4.50 or 10 pens for $7.20

d) Mobile data: 4 GB for $18 or 10 GB for $42

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D Speed, distance and time ●●○

> **Remember:** Speed = distance ÷ time. Distance = speed × time. Time = distance ÷ speed.

4. Find the speed.

a) 150 km in 2 hours **b)** 400 m in 50 seconds **c)** 100 m in 12.5 seconds

5. Find the distance travelled.

a) 60 km/h for 2.5 hours **b)** 12 m/s for 45 seconds **c)** 85 km/h for 3 hours

6. Find the time taken.

a) 240 km at 80 km/h **b)** 90 km at 60 km/h **c)** 600 m at 8 m/s

E Changing units of speed ●●●

> ⚠ **Watch out:** To change km/h to m/s, divide by 3.6. To change m/s to km/h, multiply by 3.6.

7. Change each speed to the other unit.

a) 72 km/h **b)** 15 m/s **c)** 90 km/h **d)** 10 m/s

F Problems ●●●

8. Mere drives 126 km in 1 hour 45 minutes. Find her average speed.

9. A cyclist rides at 24 km/h. How long does it take to ride 42 km? Give your answer in hours and minutes.

10. Petrol costs $2.85 per litre. A car uses 7.2 litres for every 100 km. Find the cost of petrol for a 250 km trip.

11. A 750 g box of cereal costs $6.30. A 1.2 kg box of the same cereal costs $9.60. Which is the better buy?

★ Extension

12. A train travels 180 km at 90 km/h and then 120 km at 60 km/h. Find its average speed for the whole journey.

13. Tap A fills a tank in 6 minutes. Tap B fills the same tank in 12 minutes. How long do they take to fill it together?

14. Sound travels at 340 m/s. Thunder is heard 4.5 seconds after the lightning flash. How far away is the storm, in kilometres?

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Rates and Best Buys

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

KEY WORDS rate *(pāpātanga)* · unit rate *(pāpātanga kotahi)* · best buy · speed *(tere)*

A Unit rates ●○○

Remember: A rate compares two different quantities. "Per" means "for each one".

1. Find each rate.

a) \$15 for 5 kg: cost per kg = _____

b) 280 km in 4 hours: km per hour = _____

c) \$54 for 6 hours of work: pay per hour = _____

d) 450 words in 6 minutes: words per minute = _____

e) 26 L in 4 minutes: litres per minute = _____

f) \$9.60 for 8 bottles: cost per bottle = _____

g) 468 heartbeats in 6 minutes: beats per minute = _____

h) \$4.50 for 600 g: cost per 100 g = _____

B Using a rate ●○○

2. Answer each question.

a) A tap runs at 15 litres per minute. How much water runs out in 6 minutes?

b) Hemi earns \$26.50 per hour. How much does he earn in 8 hours?

c) A car uses 7 litres of petrol for every 100 km. How much petrol does it use on a 450 km trip?

d) A printer prints 32 pages per minute. How long does it take to print 400 pages?

e) Mobile data costs \$12 per GB. Find the cost of 3.5 GB.

f) A heart beats 68 times per minute. How many times does it beat in 1 hour?

C Best buys ●●○

Remember: Find the price of one unit for each deal (the unit price), then compare.

3. Which is the better buy? Show the unit price for each.

a) Rice: 2 kg for \$6.40 or 5 kg for \$15.50

b) Juice: 1.5 L for \$4.20 or 2 L for \$5.40

c) Pens: 5 pens for \$4.25 or 12 pens for \$9.60

d) Mobile data: 5 GB for \$22 or 12 GB for \$51

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D Speed, distance and time ●●○

> **Remember:** Speed = distance ÷ time. Distance = speed × time. Time = distance ÷ speed.

4. Find the speed.

a) 180 km in 3 hours **b)** 600 m in 75 seconds **c)** 200 m in 25 seconds

5. Find the distance travelled.

a) 70 km/h for 1.5 hours **b)** 15 m/s for 40 seconds **c)** 95 km/h for 4 hours

6. Find the time taken.

a) 270 km at 90 km/h **b)** 120 km at 80 km/h **c)** 900 m at 12 m/s

E Changing units of speed ●●●

> ⚠ **Watch out:** To change km/h to m/s, divide by 3.6. To change m/s to km/h, multiply by 3.6.

7. Change each speed to the other unit.

a) 54 km/h **b)** 20 m/s **c)** 108 km/h **d)** 25 m/s

F Problems ●●●

8. Tama drives 135 km in 2 hours 15 minutes. Find his average speed.

9. A cyclist rides at 20 km/h. How long does it take to ride 45 km? Give your answer in hours and minutes.

10. Petrol costs $2.90 per litre. A car uses 6.5 litres for every 100 km. Find the cost of petrol for a 400 km trip.

11. A 500 g box of cereal costs $4.40. A 1.2 kg box of the same cereal costs $10.20. Which is the better buy?

★ Extension

12. A train travels 240 km at 80 km/h and then 100 km at 50 km/h. Find its average speed for the whole journey.

13. Tap A fills a tank in 10 minutes. Tap B fills the same tank in 15 minutes. How long do they take to fill it together?

14. Sound travels at 340 m/s. Thunder is heard 6.5 seconds after the lightning flash. How far away is the storm, in kilometres?

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